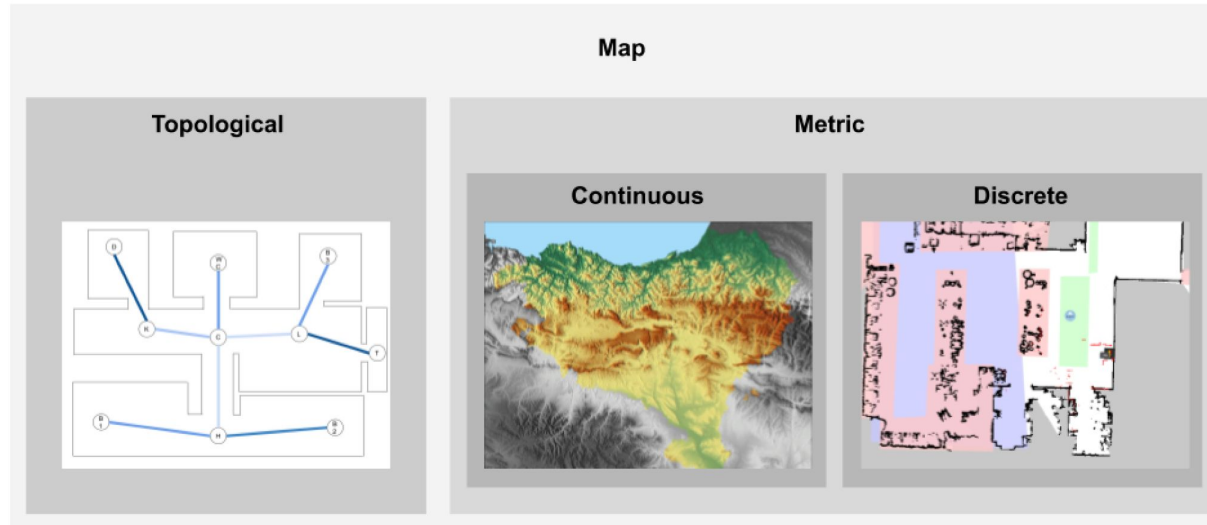


Map Representation for Robotics: Continuous Approaches and Decomposition strategies

Module 4

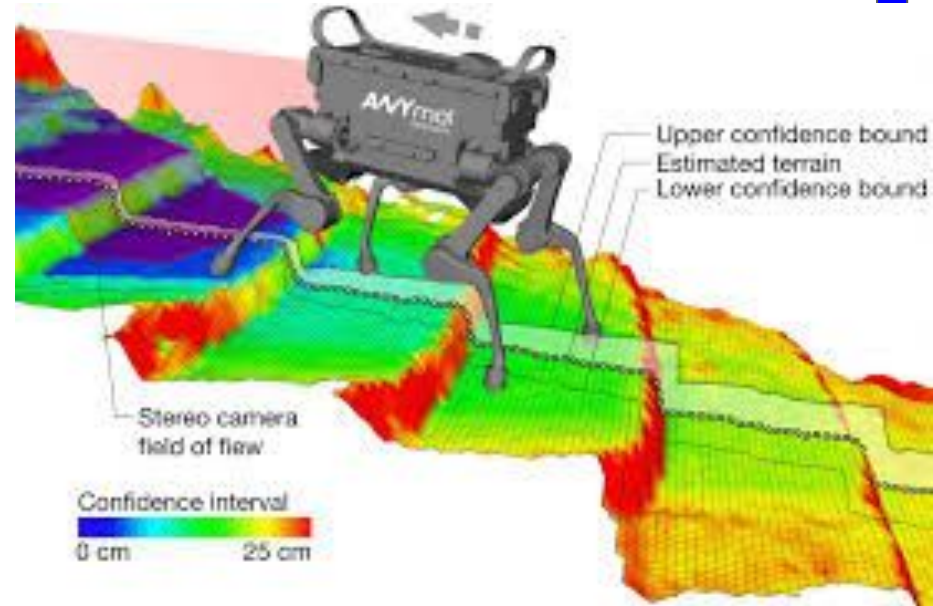
Introduction to Map Representation

- In robotics and autonomous systems, map representation is essential for navigation, localization, and path planning.
- Maps are used to model the environment in which a robot operates.
- These representations can be classified broadly into **discrete** and **continuous** representations.



Continuous Representations of Maps

- Continuous map representations model the environment as a continuous space, rather than a grid-based or discrete format.
- These representations are particularly useful for real-world applications, where the environment is not naturally divided into discrete cells.



Types of Continuous Map Representations

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1. **Geometric Representations:**

- Use mathematical models such as lines, curves, and surfaces.
- Commonly used in **Vector Maps**.
- Suitable for structured environments (e.g., road maps, building layouts).

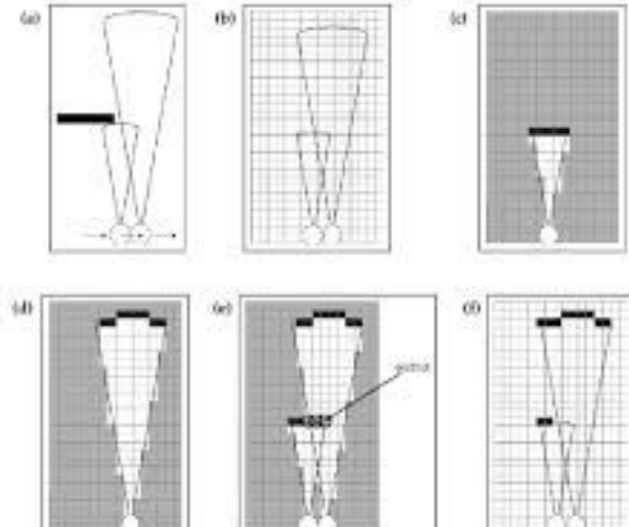
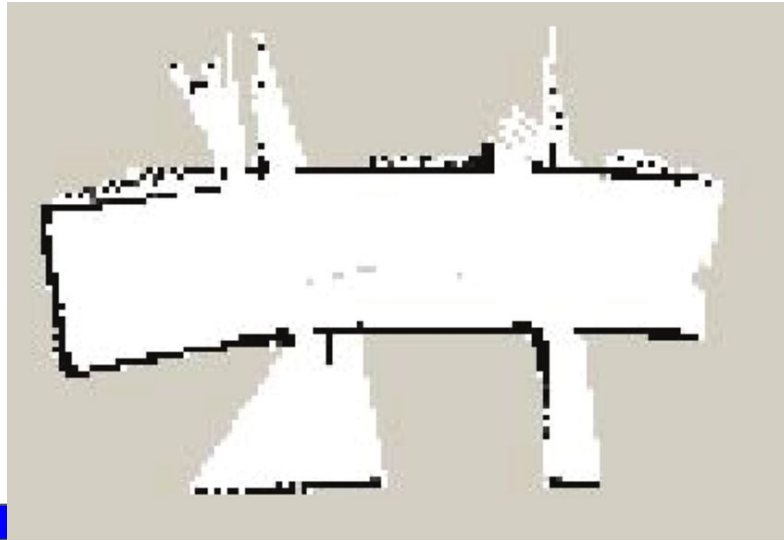
2. **Feature-Based Representations:**

- Represent an environment using distinct landmarks and features.
- Used in Simultaneous Localization and Mapping (SLAM) techniques.
- Examples: Corners, edges, walls detected using sensors.

Types of Continuous Map Representations

3. Occupancy Grid Maps (Probabilistic):

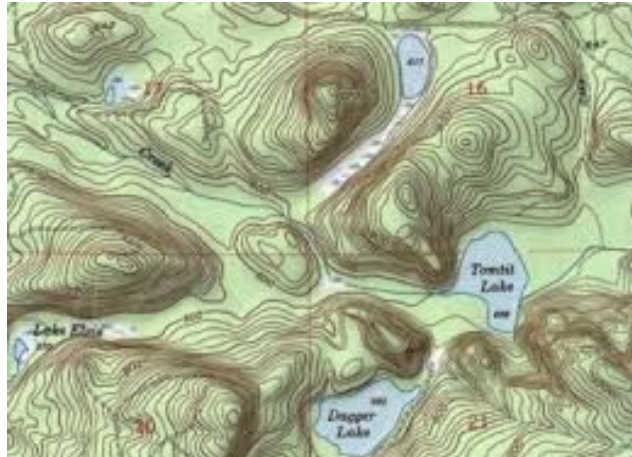
- Although often discretized, these maps maintain probability distributions, making them semi-continuous.
- Used in applications involving uncertainty, such as robotic exploration.



Types of Continuous Map Representations

4. Topological Maps:

- Represent environments as graphs where nodes indicate significant locations, and edges denote connectivity.
- Useful for high-level path planning.



Types of Continuous Map Representations

5. Signed Distance Fields (SDFs):

- Represent objects and obstacles using distance functions.
- Efficient for motion planning and obstacle avoidance.
- Used in robotics and computer graphics.

Advantages of Continuous Representations:

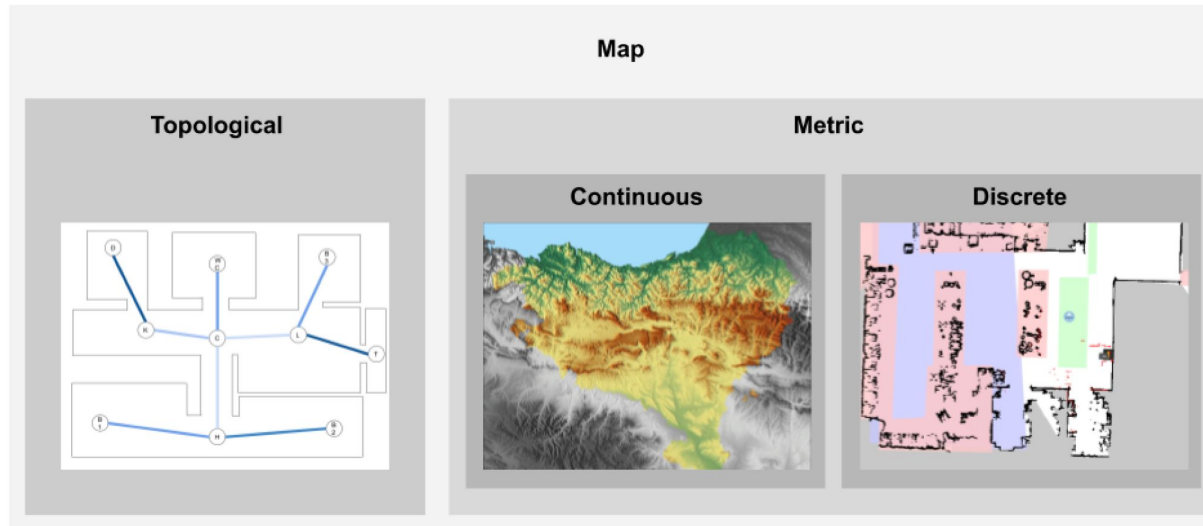
- **Higher accuracy:** Better captures real-world environments.
- **Scalability:** Can represent large areas efficiently.
- **Smooth path planning:** Avoids discretization artifacts.
- **Better handling of sensor data:** Works well with LiDAR, sonar, and camera-based perception.

Challenges of Continuous Representations:

- **Computational complexity:** Requires significant memory and processing power.
- **Difficult updates:** Modifying or dynamically updating a continuous map is challenging.
- **Uncertainty handling:** Sensor noise and inaccuracies can affect representation quality.

Decomposition Strategies in Map Representation

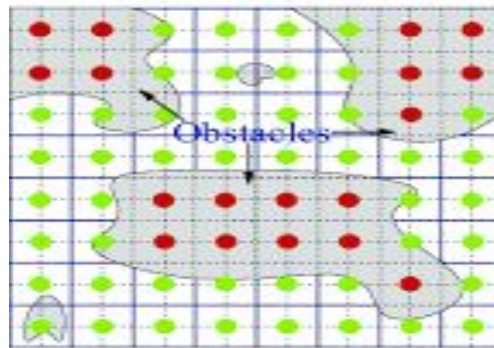
- To improve computational efficiency and manageability, environments are often decomposed into smaller, structured parts.
- **Decomposition strategies** help simplify navigation and planning problems.



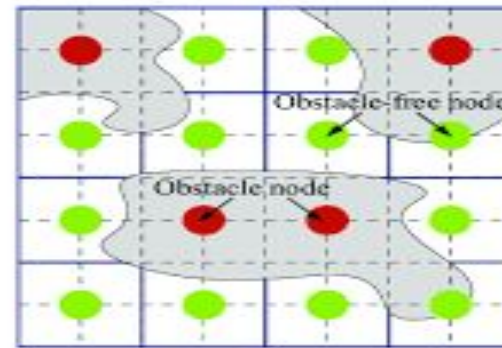
: Types of Decomposition Strategies

1. Cell Decomposition:

- Divides the environment into distinct regions or cells.
- Types:
 - **Exact Cell Decomposition:** Cells align precisely with obstacles (e.g., Voronoi Diagrams).
 - **Approximate Cell Decomposition:** Uses uniform grid-based representation (e.g., Quadtree, Octree).
- Applications: Path planning in robotics, computational geometry.



(a)

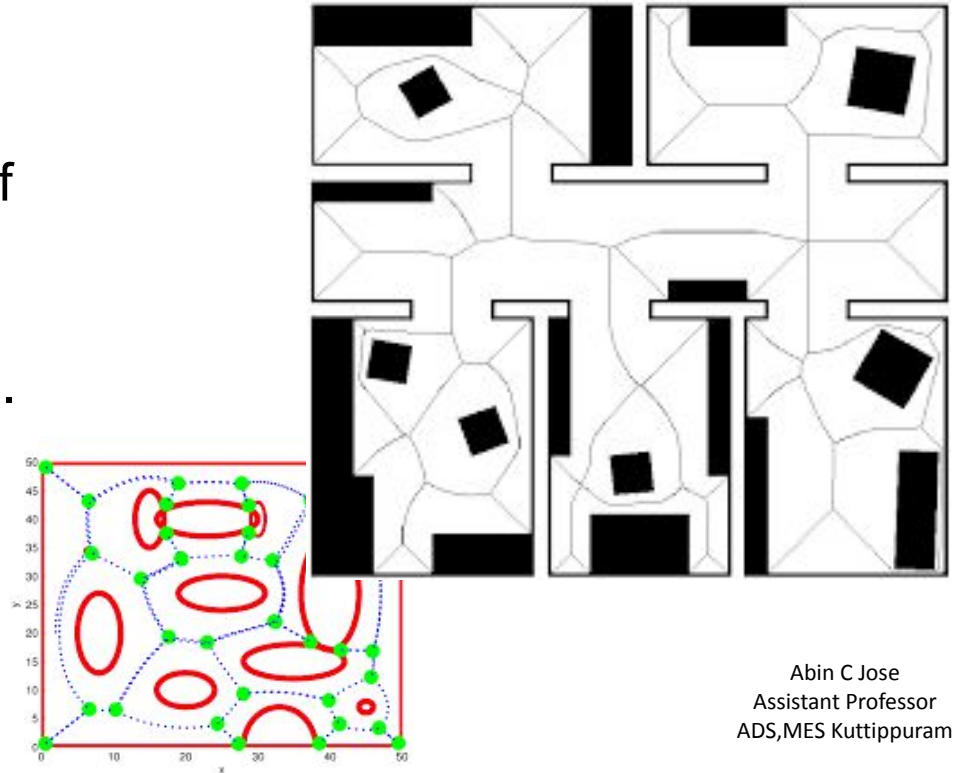


(b)

Types of Decomposition Strategies

2.Voronoi Diagram:

- Divides space based on proximity to a given set of points.
- Ensures maximum clearance from obstacles.
- Used in motion planning and autonomous navigation

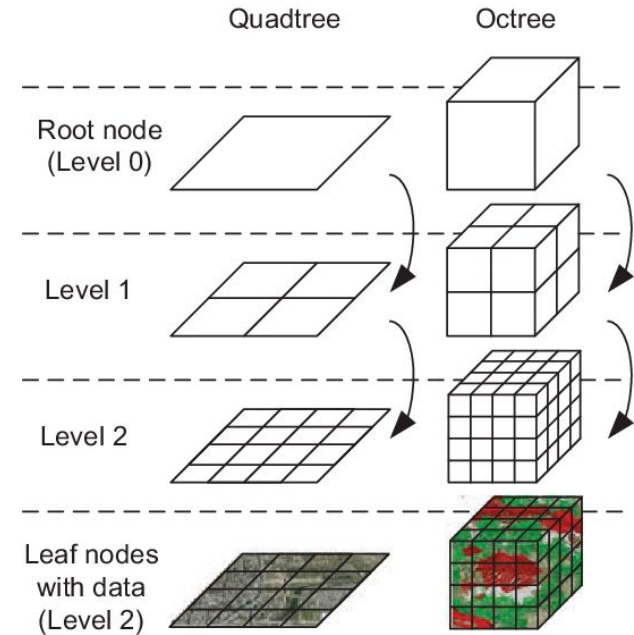


Types of Decomposition Strategies

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3. Quadtree and Octree Decomposition:

- Quadtree: A 2D space is recursively divided into quadrants.
- Octree: A 3D space is recursively divided into eight subspaces.
- Reduces memory usage and speeds up processing.
- Applied in hierarchical path planning and 3D mapping.



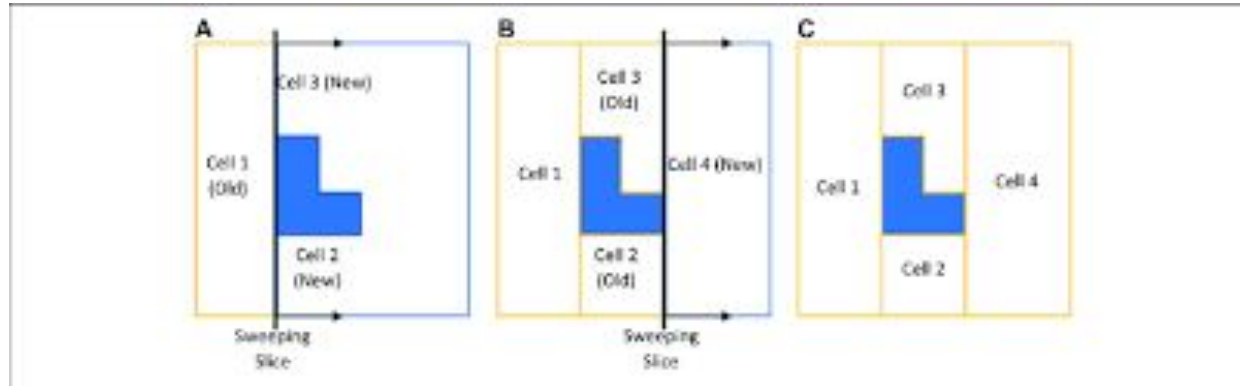
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Assistant Professor
ADS, MES Kuttippuram

Types of Decomposition Strategies

4. Boustrophedon Decomposition:

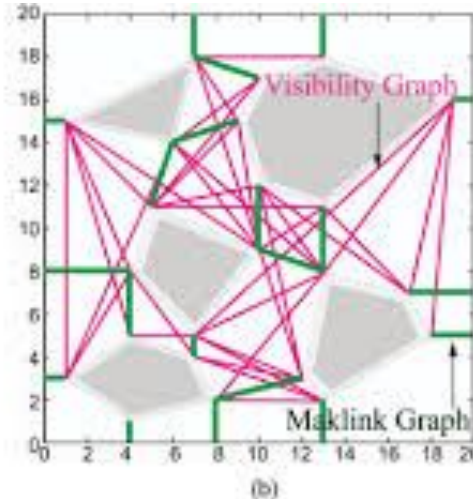
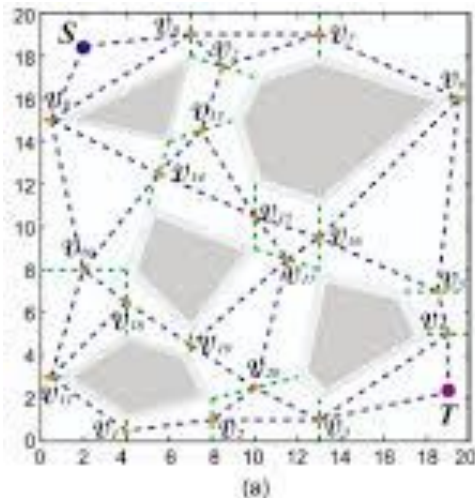
- Used for coverage path planning (e.g., robotic vacuum cleaners, agricultural robots).
- Breaks the environment into sweepable regions.



Types of Decomposition Strategies

5. Graph-Based Decomposition:

- Uses **Visibility Graphs** or **Probabilistic Roadmaps (PRM)** for efficient navigation.
- Represents connectivity between critical points in the environment.
- Suitable for high-level planning and mobile robots.



Types of Decomposition Strategies

6. Hybrid Decomposition:

- Combines multiple decomposition techniques for efficiency.
- Example: Using **Quadtrees** for coarse representation and **Voronoi Diagrams** for fine path planning.

Advantages of Decomposition Strategies:

- **Reduces computational complexity:** Breaking down the environment simplifies planning.
- **Improves real-time performance:** Efficient navigation in large-scale environments.
- **Facilitates adaptive resolution:** Allows focusing on important areas with higher detail.

Challenges:

- **Loss of precision:** Some methods approximate the environment, reducing accuracy.
- **Storage overhead:** Hierarchical methods may increase memory usage.
- **Complex implementation:** Some decomposition methods require intricate computations.